

Right-angled triangles

Acute angles that add to 90° , the hypotenuse as the longest side, and the 30° half-side rule.

LESSON 88–89

2 × 40 MIN

MATEMATIKA 7, §41

S8 5.2

LESSON CLOCK

10'

20'

25'

25'

5'

Legs and hypotenuse	10 min
The acute angles	20 min
The 30° rule	25 min
Congruence tests	25 min
Homework	5 min

LEARNING OBJECTIVES

- Name the legs and the hypotenuse and show that the hypotenuse is longest.
- Use the fact that the two acute angles add to 90° .
- Prove and apply the 30° rule: the leg opposite 30° is half the hypotenuse.
- Use the congruence tests special to right-angled triangles.

TEXTBOOK

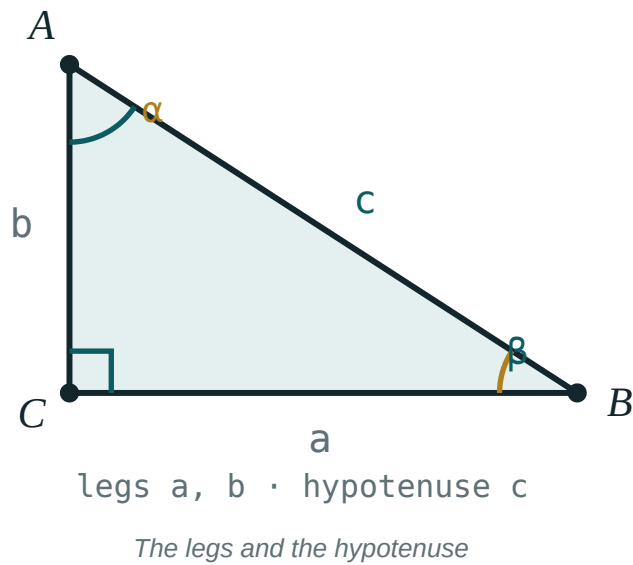
National	pp. 254–261
Cambridge	Stage 8, pp. 48–51
Workbook	Exercise 5.2

01

Explanation

Legs and hypotenuse

In a right-angled triangle the side opposite the right angle is the **hypotenuse**; the other two are the **legs**.



SIDE	OPPOSITE	LENGTH
hypotenuse	90°	the longest
each leg	an acute angle	shorter than the hypotenuse

WHY THE HYPOTENUSE MUST BE LONGEST

The larger angle faces the longer side, and the right angle is the largest of the three. The hypotenuse therefore beats both legs — before any calculation is done.

The acute angles

The three angles add to 180° and one of them is 90° , so the other two add to 90° : they are **complementary**.

$\angle A + \angle B = 90^\circ$ when $\angle C = 90^\circ$

ONE ACUTE ANGLE	THE OTHER
30°	60°
45°	45°
37°	53°
72°	18°

COMPLEMENTARY, NOT SUPPLEMENTARY

The two acute angles add to 90° , not 180° . Subtracting from 180° here is the commonest slip of the topic and always gives an impossible answer.

The 30° rule

In a right-angled triangle with an angle of 30° , the leg opposite it is exactly half the hypotenuse — and the statement runs backwards too.

$\angle A = 30^\circ \iff BC = \frac{1}{2} \cdot AB$

STATEMENT	REASON
reflect the triangle in the leg AC	construction
the two copies form a triangle with three 60° angles	$30^\circ + 30^\circ = 60^\circ$
that triangle is equilateral	equiangular $\implies$ equilateral
the doubled leg equals the hypotenuse	equal sides
$BC = \frac{1}{2} \cdot AB$	halving

HYPOTENUSE	LEG OPPOSITE 30°
10 cm	5 cm
7 cm	3.5 cm
24 cm	12 cm

THIS IS WHERE A RAMP'S RULE COMES FROM

A ramp at 30° rises one metre for every two metres of ramp. Builders quote it as a ratio; it is this theorem.

Congruence tests

Because one angle is already known to be equal, right-angled triangles need less data than others.

TEST	WHAT MUST MATCH	WHICH GENERAL CRITERION IT IS
two legs	both legs	the first — the right angle is included
a leg and an acute angle	one leg, one acute angle	the second
hypotenuse and an acute angle	hypotenuse, one acute angle	the second
hypotenuse and a leg	hypotenuse, one leg	special to right triangles

TWO ANGLES ARE NEVER ENOUGH

Knowing the right angle and one acute angle fixes the shape but not the size: the triangles are the same shape at any scale. At least one side must be given.

02**Worked examples**

EXAMPLE 1 One acute angle of a right-angled triangle is 37° . Find the other.

① The acute angles are complementary.

② $90^\circ - 37^\circ = 53^\circ$

ANSWER 53°

EXAMPLE 2 The hypotenuse is 16 cm and one angle is 30° . Find the leg opposite it.

① The leg opposite 30° is half the hypotenuse.

② $16 \div 2 = 8$

③ The leg is 8 cm .

ANSWER 8 cm

EXAMPLE 3 A leg is 6 cm and the hypotenuse is 12 cm. Find all three angles.

- 1 The leg is half the hypotenuse.
- 2 By the converse of the 30° rule, the angle opposite it is 30° .
- 3 The other acute angle is $90^\circ - 30^\circ = 60^\circ$.
- 4 The angles are $30^\circ, 60^\circ, 90^\circ$.

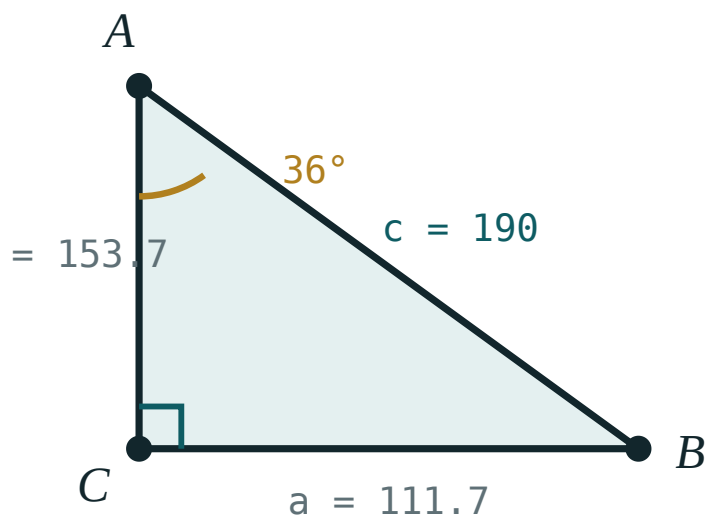
ANSWER

$30^\circ, 60^\circ, 90^\circ$

03 Interactive model

Give each pair a triangle with hypotenuse 10 cm and an angle of 30° and ask them to measure the short leg; every group reports 5 cm, and the theorem is already believed before it is proved.

The two acute angles always add to 90°

 α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Right-angled triangle	To'g'ri burchakli uchburchak	Прямоугольный треугольник
Hypotenuse	Gipotenuza	Гипотенуза
Leg	Katet	Катет
Acute angle	O'tkir burchak	Острый угол
Complementary	Yig'indisi 90° bo'lgan	Дополнительные до 90°
Opposite	Qarshi	Противолежащий
Half	Yarim	Половина
Congruence test	Tenglik alomati	Признак равенства

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The side opposite the right angle is the:

leg

hypotenuse

median

base

The hypotenuse is:

the shortest side

the longest side

equal to a leg

any side

The two acute angles add to:

45°

90°

180°

360°

One acute angle is 72° ; the other is:

18°

28°

108°

118°

The leg opposite 30° is:

the hypotenuse

half the hypotenuse

twice the hypotenuse

equal to the other leg

Two right-angled triangles with two equal angles are:

congruent

not necessarily congruent

equal in area

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. One acute angle 37° : the other

ANSWER 53°

2. One acute angle 45° : the other

ANSWER 45°

3. One acute angle 72° : the other

ANSWER 18°

4. Hypotenuse 10 cm , angle 30° : the opposite leg

ANSWER 5 cm

5. Hypotenuse 7 cm , angle 30° : the opposite leg

ANSWER 3.5 cm

6. The longest side of a right-angled triangle

ANSWER the hypotenuse

7. Name the sides beside the right angle

ANSWER the legs

Medium · the standard question

7 PROBLEMS

1. Leg 6, hypotenuse 12: the angles

ANSWER $30^\circ, 60^\circ, 90^\circ$

2. Hypotenuse 24 cm, angle 60° : the shorter leg

ANSWER 12 cm

3. One acute angle is twice the other: find both

ANSWER 30° and 60°

4. One acute angle is 10° more than the other

ANSWER 40° and 50°

5. Can a right-angled triangle be obtuse-angled?

ANSWER No

6. Can a right-angled triangle be isosceles?

ANSWER Yes — the acute angles are 45° each

7. A ramp at 30° is 6 m long: its rise

ANSWER 3 m

Hard · stretch and reason

7 PROBLEMS

1. One acute angle is 4 times the other: find both

ANSWER 18° and 72°

2. The hypotenuse is 2 times a leg: find the angles

ANSWER 30° , 60° , 90°

3. A leg is 9 cm and it faces 30° : find the hypotenuse

ANSWER 18 cm

4. Which test lets two right triangles be congruent from a hypotenuse and a leg?

ANSWER The test special to right-angled triangles

5. Prove the 30° rule by reflection

ANSWER The two copies form an equilateral triangle

6. The acute angles are $2x$ and $3x$: find them

ANSWER 36° and 54°

7. Why are two equal angles not enough for congruence here?

ANSWER The size is not fixed — only the shape

07 Homework

Homework — 5 tasks

Subtract from 90° , never from 180° , when finding the second acute angle.

1. One acute angle is 28° . Find the other.
2. The hypotenuse is 18 cm and one angle is 30° . Find the opposite leg.
3. A leg is 7 cm and the hypotenuse 14 cm. Find all three angles.
4. The acute angles are in the ratio 1 : 5. Find them.
5. List the four congruence tests for right-angled triangles and say which general criterion each is.